

Architecture Specification: CXS Bridge Encryption and Data Management System

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Project: CXS Bridge Encryption Project

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1. Introduction

1.1 Purpose

This document defines the architecture of the CXS Bridge Encryption and Data Management System, a hardware module designed for secure data transfer between a CXS interface and multiple internal logic blocks. The bridge provides:

- Configurable data encryption.
- Parity generation for data integrity.
- Centralized data storage.
- Configuration management.
- Interface arbitration between software and internal hardware blocks.

The design targets a VLSI implementation and is intended to operate as an intermediate communication bridge between external software and internal processing modules.

2. System Overview

The CXS Bridge receives configuration packets and data packets from the CXS interface, encrypts incoming payloads using a configurable expansion algorithm, appends parity information, and stores the processed data inside the Central Data

Memory (CDM). Multiple internal logic blocks access the CDM using dedicated address regions assigned during system configuration.

The Control Unit manages:

- System initialization and configuration registers.
- Phase transitions (Configuration, Link-Up, Data Transmission).
- Memory access arbitration and status reporting.

3. High-Level Architecture

The system consists of five major functional blocks:

1. **CXS Interface:** The gateway for external communication.
2. **Control Unit & Register File:** The system coordinator and configuration repository.
3. **Encryption Module:** Bit-level expansion logic.
4. **Parity Module:** Error-detection logic.
5. **Central Data Memory (CDM):** Unified storage for processed data.

4. Functional Modules

4.1 CXS Interface

The primary communication endpoint responsible for:

- Receiving configuration and data packets.
- Sending status responses and flow control.
- Header validation and packet framing.

For more information on the CXS protocol, refer to the [Arm CXS Specification](#).

4.2 Control Unit

Coordinates the overall state machine and CDM arbitration. It transitions through three main phases:

1. **Configuration:** Registration and mode selection.
2. **Link-Up:** Header validation and target identification.
3. **Data Transmission:** Payload processing and storage.

4.3 Encryption Module

Performs a configurable bit-expansion algorithm based on the parity of input words.

Rule Component	Description
Step 1: Bit Counting	Counts number of logic '0' and '1' bits in the input word.
Step 2: Mode A (Even)	If count is even, bits are duplicated/appended (e.g., 1011 becomes 11001111).
Step 3: Mode B (Odd)	Expansion is triggered if the selected bit count is odd.

Note:- If the counts of 0s and 1s are equal, or if both counts are even, the expansion shall map '0' to '01' and '1' to '10' (e.g., input '10' becomes '1001').

4.4 Parity Module

Generates a parity bit (Even or Odd) which is appended to the encrypted data before CDM storage to ensure data integrity.

4.5 Central Data Memory (CDM)

The internal storage repository for encrypted payloads, headers, and parity bits. Each logic block is restricted to its assigned memory region to prevent data conflicts.

4.6 Address Translation Unit

The Address Translation Unit is responsible for managing memory mapping and operates in two primary modes:

1. Limit Address and Last Address Mode.

2. Region Mode, which defines specific address ranges (e.g., region 1 spanning from address 10 to address 50).

The address section for this unit consists of a combination of the region number and an offset.

5. Register File Definitions

Base Address: 0x0000

Bits	Field	Description
31:17	RSV	Reserved for future use
16	ADDR_MODE	0- > address range 1- > region mode
15:8	DEV_COUNT	Number of connected devices
7:4	ENC_MODE	Encryption mode selection (0000: Even, 0001: Odd)
3:2	PARITY_MODE	Parity configuration
1:0	SYS_STATE	System operating state (Idle, Config, Link-Up, Transfer)

Device Index	Base address	description
Device 1	determine by software	Start address [7:0] - End address [8:15]

Device Index	Base address	description
Device 2	determine by soft ware	Start address [7:0] - End address [8:15]
Device 3	Determine by software	Start address [7:0] - End address [8:15]

6. System Operation Phases

Phase 1: Configuration

Software sends a configuration packet. The Control Unit stores registers, defines device counts, selects the encryption mode, and allocates address space for logic blocks.

Phase 2: Link-Up

Software transmits a Header packet. The bridge validates the header and identifies target logic. Successful validation returns a **STATUS = VALID** to proceed to Data Transmission.

Phase 3: Data Transmission

The payload is received, encrypted, assigned parity, and stored in the CDM. The destination logic is notified upon completion.

7. Technical Specifications

7.1 CXS Interface Signals

Signal Name	Width	Direction	Description
TX_DATA	16-bits ▾	Output ▾	Data transmit bus

Signal Name	Width	Direction	Description
RX_DATA	16-bits ▾	Input ▾	Data receive bus
TX_VALID	1-bit ▾	Output ▾	Indicates valid transmit data
RX_VALID	1-bit ▾	Input ▾	Indicates valid receive data
READY	1-bit ▾	Output ▾	Flow control signal
CLK	1-bit ▾	Input ▾	System clock
RST_N	1-bit ▾	Input ▾	Active-low reset

7.2 Transaction Layer Packet (TLP)

Word	Description	Format
Word 0	Header	Described in register header table
Word 1	Payload 0	Address [15:8], Data [7:... ▾
Word 2	Payload 1	Address [15:8], Data [7:... ▾
Word 3	Payload 2	Address [15:8], Data [7:... ▾

8. Error Handling & Future Enhancements

The bridge detects invalid headers, unsupported encryption modes, parity mismatches, and invalid state transitions, reporting them through dedicated status registers.

Future Roadmap:

- AES-based encryption engine.
- ECC protection (replacing single-bit parity).
- DMA support and variable-length TLP packets.
- CRC-based packet verification.

9. Clocking Architecture

The CXS Bridge operates using two independent clock domains.

- **CXS Clock Domain (CLK_CXS):** CXS Interface, RX/TX Logic, Packet Parser.
- **System Clock Domain (CLK_SYS):** Control Unit, Register File, Encryption Module, Parity Module, CDM, Internal Logic Blocks.

The CXS Interface is intentionally isolated from the remainder of the system and operates on an independent clock (CLK_CXS). All internal processing logic operates using the system clock (CLK_SYS). This separation allows the bridge to interface with external devices operating at different frequencies while maintaining deterministic timing for the internal processing pipeline.

10. Clock Domain Crossing (CDC)

Since the CXS Interface and the internal processing logic operate in different clock domains, all signals crossing between CLK_CXS and CLK_SYS shall use dedicated Clock Domain Crossing (CDC) circuitry.

10.1 CDC Overview

The CXS Interface and Packet Decoder reside in the CLK_CXS domain, while the Control Unit, Encryption Engine, Parity Generator, and CDM reside in the CLK_SYS domain. They are separated by a CDC boundary consisting of an Asynchronous FIFO.

10.2 Data Transfer

Packet data transferred from the CXS Interface to the Control Unit shall pass through an Asynchronous FIFO. The FIFO provides safe clock-domain crossing, elastic buffering, rate matching, back-pressure handling, and metastability protection. It shall implement independent read and write clocks, Gray-coded read/write pointers, Full and Empty detection, and pointer synchronization using dual flip-flops.

10.3 Control Signal Synchronization

Single-bit control signals crossing clock domains shall be synchronized using a two-stage synchronizer. Typical synchronized signals include Packet Available, FIFO Empty, FIFO Full, Ready, Link-Up Complete, Configuration Complete, and Error Status.

10.4 Multi-Bit Signals

Multi-bit buses shall be transferred using mechanisms such as Asynchronous FIFO (preferred), Valid/Ready handshake protocol, or Request/Acknowledge handshake to prevent data incoherency.

10.5 CDC Design Rules

- No combinational logic between synchronizer stages.
- No fan-out from the first synchronizer flip-flop.[The output of the **first synchronizer flip-flop** shall drive **only the input of the second synchronizer flip-flop**.]
- All asynchronous interfaces shall be verified using static CDC analysis.
- All asynchronous FIFOs shall be formally verified.

11. Reset Strategy

The bridge implements a global reset distributed to both clock domains.

11.1 Reset Signals

- **RST_N_CXS (CLK_CXS):** Active-low reset for the CXS Interface.
- **RST_N_SYS (CLK_SYS):** Active-low reset for all internal logic.

11.2 Reset Assertion

Reset assertion is asynchronous. When reset is asserted, the Control Unit returns to Idle, the Register File is cleared, CDM write operations are disabled, encryption and parity modules are reset, FIFOs are flushed, and status registers are cleared.

11.3 Reset Deassertion

Reset deassertion shall be synchronous to each clock domain using a reset synchronizer (dual flip-flops) to prevent metastability during reset release.

11.4 Power-Up Sequence

1. Reset asserted.
2. Internal clocks become stable.
3. Reset released independently in each clock domain.
4. FIFO pointers initialized.
5. Register File initialized.
6. Control Unit enters Configuration state.
7. Software begins bridge configuration.

11.5 Recommended Reset Values

- **Control Unit:** Idle
- **Register File:** Default Values
- **CDM:** Undefined
- **Encryption/Parity Modules:** Disabled
- **FIFO:** Empty
- **Status Register:** Cleared